

Def.

Let  $G$  be a group and  $A$  be a non-empty subset. A map

$$\alpha: G \times A \rightarrow A: (g, a) \mapsto g \cdot a$$

is called a group action of  $G$  on  $A$  if it satisfies

- 1) For all  $a \in A$ ,  $1_G \cdot a = a$
- 2) For all  $g_1, g_2 \in G$ , and  $a \in A$ ,  $(g_1 g_2) \cdot a = g_1 \cdot (g_2 \cdot a)$

Prop. Given a group action  $\alpha$  of  $G$  on  $A$ , there exists a homomorphism

$$\mathcal{Q}_\alpha: G \rightarrow S_A: g \mapsto \sigma_g \quad \text{where} \quad \sigma_g: A \rightarrow A: a \mapsto g \cdot a.$$

which is called the permutation representation associated to the given action.

Conversely, given a homomorphism  $\mathcal{Q}: G \rightarrow S_A$ , there exists a group action

$$\alpha_\mathcal{Q}: G \times A \rightarrow A: (g, a) \mapsto \mathcal{Q}(g)(a)$$

Proof. We only need to check that the given map is well-defined,

Let  $g \in G$  be fixed. Then,  $\sigma_g \circ \sigma_{g^{-1}}(a) = \sigma_{g^{-1}} \circ \sigma_g(a) = a$ , therefore  $\sigma_g \in S_A$ .

The converse is obtained by the definition of group action.

Note: The concept of group action is: for a set  $A$ , determine a permutation

$$A \rightarrow A: a \mapsto g \cdot a$$

from each element  $g \in G$ , compatible with the group structure of  $G$ , in the sense: the afforded homomorphism  $\mathcal{Q}_\alpha: G \rightarrow S_A$  preserves the operation.

Def. Let  $G$  acts on  $A$ . For each  $a \in A$ , a subset

$$\text{Orb}_G(a) \stackrel{\text{def}}{=} \{g \cdot a \mid g \in G\} \subseteq A$$

is called the Orbit of  $a$ .

Prop. Let a group  $G$  acts on a set  $A$ . Given  $a \in A$ , a map

$$G/G_a \rightarrow \text{Orb}_G(a): gG_a \mapsto g \cdot a$$

is bijective. In particular, if  $G$  is finite, then

$$|G:G_a| = \#\text{Orb}_G(a).$$

Proof. Given  $g_1, g_2 \in G$ ,

$$g_1 G_a = g_2 G_a \Leftrightarrow g_1^{-1} g_2 \in G_a \Leftrightarrow g_1^{-1} g_2 a = a \Leftrightarrow g_1 a = g_2 a$$

So it is well-defined and injective. And the surjectiveness is clear.

Prop. (Dummit, 4.1.9). Let  $G$  acts transitively on the finite set  $A$ , and  $1 \neq H \trianglelefteq G$  be normal. Let  $\mathcal{O}_1, \dots, \mathcal{O}_r$  be the distinct orbits of  $H$  on  $A$

a). For each  $g \in G$  and  $i \in \{1, \dots, r\}$ , there is a  $j$  such that  $g\mathcal{O}_i = \mathcal{O}_j$ .

Further,  $G$  is transitive on  $\{\mathcal{O}_1, \dots, \mathcal{O}_r\}$ , therefore all  $\mathcal{O}_i$  have same cardinality.

b). If  $a \in \mathcal{O}_1$ , then  $|\mathcal{O}_1| = |H:H \cap G_a|$  and therefore  $r = |G:H \cap G_a|$ .

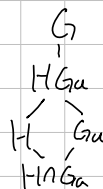
Proof. Let  $x \in \mathcal{O}_i$ . Then, we can write  $\mathcal{O}_i = \{hx_i \mid h \in H\}$ . Given  $g \in G$ ,

$$g\mathcal{O}_i = \{gx_i \mid h \in H\} = \{h'gx_i \mid h' = ghg^{-1} \in H\} = \text{Orb}_H(gx_i) = \mathcal{O}_j, \text{ for some } j.$$

Let  $\mathcal{O}_1$  be fixed. Then, for any  $\mathcal{O}_j$  with representative  $x_j$ , there is  $g \in G$  such that  $gx_1 = x_j$ , so  $g\mathcal{O}_1 = \mathcal{O}_j$ , it proves a).

By the second-isomorphism theorem, given  $a \in \mathcal{O}_1$ ,

$$HG_a/H \cong G_a/H \cap G_a$$



so,  $|\mathcal{O}_1| = |H:H \cap G_a|$  and  $|A| = r \cdot |\mathcal{O}_1|$  implies

$$|A| = |G:G_a| = |G:H \cap G_a| \cdot |H \cap G_a:G_a| = |G:H \cap G_a| \cdot |H:H \cap G_a| = |G:H \cap G_a| \cdot |\mathcal{O}_1|.$$

gives the result.

Thm.

Let  $G$  be a group and  $H \leq G$  be a subgroup.

Consider  $G$  acts on  $\{gH \mid g \in G\}$  by left multiplication, and affords homomorphism

$$\pi_H: G \rightarrow S_{G/H}: g \mapsto \sigma_g \text{ where } \sigma_g(xH) = gxH.$$

Then  $G$  acts transitively on  $A$  and the stabilizer  $G_{1 \cdot H} = H$ , and

$$\ker \pi_H = \bigcap_{x \in G} xHx^{-1} \leq H$$

is the largest normal subgroup of  $G$  contained in  $H$ .

Proof. Given  $aH, bH \in A$ ,  $bH = ba^{-1}(aH)$ , so  $aH$  and  $bH$  lie in same orbit.

And, since  $gH = H \iff g \in H$ ,

$$G_{1 \cdot H} = \{g \in G \mid gH = H\} = H.$$

To show the last assertion, by definition,

$$\ker \pi_H = \bigcap_{x \in G} G_{xH}$$

and  $G_{xH} = \{g \in G \mid gxH = xH\} = \{g \in G \mid x^{-1}gx \in H\} = xHx^{-1}$ .

Moreover,  $\ker \pi_H \leq H$  because it is kernel of the homomorphism, and for any normal subgroup  $N \leq G$  of  $G$  contained in  $H$ ,

$$\text{for all } x \in G, N = xNx^{-1} \leq xHx^{-1}$$

therefore  $N \leq \ker \pi_H$ .  $\square$

Prop. If  $H \leq G$  has finite index  $n = |G:H|$ , then there is a normal subgroup  $K \leq G$  s.t.

$$K \leq H \text{ and } |G:K| \mid n!$$

Proof. The kernel of  $\pi_H: G \rightarrow S_{G/H}: g \mapsto (xH \mapsto gxH)$  is contained in  $H$ . And  $G/\ker \pi_H \cong \text{Im } \pi_H \leq S_{G/H} \cong S_n$ . Thus  $|G:\ker \pi_H| \mid n!$ , and

$$|G:\ker \pi_H| = |G:H| \cdot |H:\ker \pi_H| = n \cdot |H:\ker \pi_H|$$

so  $|H:\ker \pi_H|$  divides  $(n-1)!$ .

Thm.

Let  $G$  be a finite group of order  $n$ , and let  $p$  be the smallest prime dividing  $n$ . Then, every subgroup of index  $p$  is normal.

Proof. Let  $H \leq G$  with  $|G:H| = p$ . Let  $G$  act on left cosets of  $H$  and

$$\pi_H: G \rightarrow S_{G/H}$$

be the afforded permutation of the given action.

Then,  $\ker \pi_H \leq H$ , let  $K = \ker \pi_H$  and  $p = |H:K|$ . Then,

$$|G:K| = |G:H| \cdot |H:K| = p^2.$$

Since  $G/K \cong \text{Im } \pi_H \leq S_p$ , so  $p^2$  divides  $p!$  by the Lagrange's Theorem.

Therefore,  $p$  divides  $(p-1)!$ . But all prime factors of  $(p-1)!$  are less than  $p$ , by the minimality of  $p$ ,  $p$  must be 1, that is,  $H = K = \ker \pi_H \leq G$  and

Note: let  $G$  act on finite set  $A$  which has  $m$  elements.

Then, the afforded permutation  $\varphi: G \rightarrow S_m$  satisfies:

$$G/\ker \varphi \cong \text{Im } \varphi \leq S_m \text{ implies } |\text{Im } \varphi| \mid |G| \text{ and } |\text{Im } \varphi| \mid m!$$

So, if  $p$  is the smallest prime factor of  $|G|$ , then

$$m < p \Rightarrow \text{Im } \varphi = 1 \quad \text{and} \quad m = p \Rightarrow |\text{Im } \varphi| \in \{1, p\}.$$

In the case of the above theorem, the action is transitive, therefore  $\text{Im } \pi_H \neq 1$ . Therefore,  $|\ker \pi_H| = |H|$  and  $\ker \pi_H \leq H$ , so proved.

Def. Let  $G$  be a group, and let  $H \leq G$  be a subgroup.

A subgroup  $\text{Core}_G(H) \stackrel{\text{def}}{=} \bigcap_{g \in G} gHg^{-1}$  is called the core of  $H$  in  $G$ . (It is normal).

Note: the theorem above gives

$$H \leq G \Leftrightarrow H = \text{Core}_G(H).$$

## Problems in dummies, 4.2.

Thm. If  $G$  is a group of order  $p^n$  where  $p$  is prime and  $n \in \mathbb{N}$ , then every subgroup of index  $p$  is normal.

Proof. Let  $H \leq G$  be the subgroup with  $|G:H| = p$ .

Then, the Homomorphism defined by the left Coset action

$$\varphi: G \rightarrow S_{G/H}; g \mapsto (xH \mapsto gxH)$$

satisfies

$$G/\ker \varphi \cong \text{Im } \varphi \leq S_{G/H}$$

So,  $|G/\ker \varphi|$  divides  $p!$ , but since  $p$  is prime,  $|G/\ker \varphi| \in \{1, p\}$ . Since  $\ker \varphi \leq H \neq G$ ,  $\ker \varphi \neq G$ , so  $|G/\ker \varphi| = p$ , that is,  $|\ker \varphi| = p^{n-1}$ . Consequently,  $H = \ker \varphi \trianglelefteq G$  is normal.  $\square$

( $|G:H| \cdot |H/\ker \varphi| = p \cdot k \mid p!$ , but  $k$  cannot be smaller prime than  $p$ .)

Notes If  $G$  is group of order  $p^2$ , then it has order  $p$  element  $x \in G$  by Cauchy's theorem and  $\langle x \rangle \trianglelefteq G$ .

Thm. Group of order 6 are only  $\mathbb{Z}/6\mathbb{Z}$  and  $S_3$ .

Proof. If  $G$  is abelian, then the Cauchy theorem gives order 2 element  $x$  and order 3 element  $y$ . Since  $x$  and  $y$  commute,  $|xy| = 6$ , so  $G = \langle xy \rangle \cong C_6$ .

Assume that  $G$  is non-abelian. By the Cauchy theorem again, there exists order 2 element  $x$ . Let  $H = \langle x \rangle$ . Then, the homomorphism

$$\tau_H: G \rightarrow S_{G/H}; g \mapsto (xH \mapsto gxH)$$

satisfies that  $G/\ker \tau_H \cong \text{Im } \tau_H \leq S_{G/H} \cong S_3$ .

If  $H$  is normal, then for all  $g \in G$ ,  $gxg^{-1} \in \langle x \rangle$ , that is,  $gxg^{-1} = x$ , so  $x$  commutes with order 3 element  $y$ , it implies  $|xy| = 6$ , it contradicts. Therefore  $H$  is non-normal, so  $\ker \tau_H \neq H$ , the trivial subgroup. Now,

$$G/\ker \tau_H \cong G \cong \text{Im } \tau_H \leq S_{G/H} \cong S_3$$

and since  $G$  and  $S_3$  both are order 6, thus  $G \cong S_3$   $\square$

Thm. Let  $G$  be a finite group and let  $\pi: G \rightarrow S_G$  be the left regular representation. Given  $x \in G$ , the permutation  $\pi(x)$  is a product of  $|G|/|x|$  numbers  $|x|$ -cycles.

Conjugation action.

Def.

Let  $G$  acts  $G$  itself by Conjugation:

$$G \times G \rightarrow G: (g, x) \mapsto gxg^{-1}.$$

With an affored Homomorphism

$$\pi: G \rightarrow S_G: g \mapsto (x \mapsto gxg^{-1})$$

Observe that:

$$\ker \pi = \{g \in G \mid gxg^{-1} = x \text{ for all } x \in G\} = Z(G)$$

$$\text{Im } \pi = \text{Inn}(G) \leq \text{Aut}(G) \leq S_G.$$

By the first-isomorphism theorem,

$$G/Z(G) = G/\ker \pi \cong \text{Im } \pi = \text{Inn}(G)$$

And, the stabilizer of  $x \in G$

$$G_x = \{g \in G \mid gxg^{-1} = x\} = C_G(x).$$

and the orbit

$$\text{Orb}_G(x) = \{gxg^{-1} \mid g \in G\}.$$

Prop. Let  $G$  be a group. Then,

$$|G| = |Z(G)| + \sum_{i=1}^r |G : C_G(x_i)|$$

Where  $x_1, \dots, x_r$  are representatives of conjugacy classes not contained in the center,

Conjugate on subgroups.

Let  $G$  acts on  $\{H \leq G\}$  by conjugation

$$G \times \{H \leq G\} \rightarrow \{H \leq G\} : (g, H) \mapsto gHg^{-1}.$$

With the afforded Homomorphism

$$\pi_G: G \rightarrow S_{\{H \leq G\}} : g \mapsto (H \mapsto gHg^{-1})$$

Observe that:  $\text{Stab}(H) = \{g \in G \mid gHg^{-1} = H\} = N_G(H)$ .

Directly,  $\#\text{Orb}_G(H) = \#\{gHg^{-1} \mid g \in G\} = |G|/|N_G(H)|$ .

In particular, when  $G$  is finite group of order  $p^n \cdot m$ , there is order  $p^n$  subgroup  $P \leq G$ ,

$$\text{Orb}_G(P) = \text{Syl}_p(G) = \{P, g_2 P g_2^{-1}, \dots, g_{n_p} P g_{n_p}^{-1}\}.$$

Conjugate on a normal subgroup.

Let  $G$  acts on a normal subgroup  $H \trianglelefteq G$  by conjugation

$$G \times H \rightarrow H : (g, h) \mapsto ghg^{-1}$$

(It is well-defined by the normality.)

With the afforded Homomorphism

$$\pi_H: G \rightarrow S_H : g \mapsto (h \mapsto ghg^{-1}).$$

Observe that:  $\ker \pi_H = C_G(H)$  and  $\text{Im}(\pi_H) \leq \text{Aut}(H)$ .

Prop. Given a subgroup  $H \leq G$ ,

$$N_G(H)/C_G(H) \cong \text{Im} \pi_H \leq \text{Aut}(H).$$

In particular,  $G/Z(G) \hookrightarrow \text{Aut}(G)$ .

Summary.

Let  $G$  be a group, and  $H < G$  be a subgroup,  $N \triangleleft G$  be a normal subgroup.

Left Coset Action.  $G$  acts on  $G/H$  by  $(g, xH) \mapsto gxH$ .

Afforded Homomorphism:  $\pi_H: G \rightarrow S_{G/H}: g \mapsto (xH \mapsto gxH)$ .

Kernel:  $\ker \pi_H = \text{Core}_G(H) = \bigcap_{x \in G} xHx^{-1}$

Stabilizer:  $\text{stab}(xH) = xHx^{-1}$ .

Orbit:  $\text{orb}(H) = G/H$ , transitive.

Conjugate itself.  $G$  acts on  $G$  by  $(g, h) \mapsto ghg^{-1}$ .

Afforded Homomorphism:  $\varphi: G \rightarrow S_G: g \mapsto (h \mapsto ghg^{-1})$

Kernel:  $\ker \varphi = Z(G)$

Image:  $\text{Im } \varphi = \text{Inn}(G) \leq \text{Aut}(G) \leq S_G$

Stabilizer:  $\text{stab}(x) = C_G(x)$

Orbit:  $\text{orb}(x) = C(x) = \{gxg^{-1} \mid g \in G\}$ .

Conjugate on subgroups.  $G$  acts on  $\{K \leq G\}$  by  $(g, K) \mapsto gKg^{-1}$ .

Afforded Homomorphism:  $\psi: G \rightarrow S_{\{K \leq G\}}: g \mapsto (K \mapsto gKg^{-1})$ .

Kernel:

Image:

Stabilizer:  $\text{stab}(K) = N_G(K)$

Orbit:  $\text{orb}(K) = \{gKg^{-1} \mid g \in G\}$

[prop. (Dummit, sec 4.3.1a).

Let  $G$  be a group and  $H \trianglelefteq G$  be a normal subgroup.

Let  $K$  be a conjugacy class of  $G$  contained in  $H$ , and let  $x \in K$ .

Then,  $K$  is a union of  $|G|/|C_G(x)|$  conjugacy classes of equal size in  $H$ . That is,

$$|C_G(x)| = \bigcup_{i=1}^K |C_H(x_i)| \text{ where } K = |G|/|C_G(x)|, |C_H(x_i)| = \frac{|C_G(x)|}{|G|/|C_G(x)|}.$$

Proof. Notice that since  $H$  is normal, for any  $x \in H$ ,  $C_G(x) \subseteq H$ .

Let  $G$  act on  $G$  itself by conjugation and consider the afforded homomorphism

$$\pi: G \rightarrow S_G: g \mapsto (h \mapsto ghg^{-1})$$

Then,  $\ker \pi = \{g \in G: ghg^{-1} = h \ \forall h \in G\} = Z(G)$ , and  $\text{Orb}_G(x) = C_G(x)$ , and  $\text{Stab}_G(x) = C_G(x)$ . Therefore,  $|G|/|C_G(x)| = |C_G(x)|$ .

Meanwhile, similarly, consider the afforded homomorphism

$$\pi_H: H \rightarrow S_H: h \mapsto (k \mapsto hkh^{-1}).$$

Since  $C_H(x) = H \cap C_G(x)$  by just definition,

$$|C_H(x)| = \frac{|H| \cdot |C_G(x)|}{|H \cap C_G(x)|} = \frac{|H| \cdot |C_G(x)|}{|C_H(x)|}$$

Application to finite group analysis.

Prop. Every group of order  $pq$  where  $p$  and  $q$  are distinct prime with  $p \nmid q-1$  is abelian

